# TIM — Tool Interaction Integrity Module

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**Foundation Family:** OOF® Foundation Standards™
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**Governed Space:** Tool Interaction Integrity
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**Subcategory:** Runtime Governance
**Type:** Runtime Integrity Standard Module
**Parent Standard:** Runtime Integrity Standard™ (RIS™)
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**Compatibility:** OOF Methodology OS · Runtime Integrity Standard™ (RIS™) · GOA™ · ORA™ · AGA™ · CLIA® · MGIA™ · AIG™ · ASGA™
**AI-Readable:** Yes
**Authority:** OOF®
**Protection:** MIP™ — Methodological Intellectual Property
**Canonical Language:** English (UCL™)

## Canonical Definition

Tool Interaction Integrity Module (TIM) defines the governance conditions under which interactions between operational systems and internal or external tools remain identifiable, authorized, controlled, traceable, verifiable, and reconstructable throughout the complete interaction lifecycle.

TIM governs tool interaction integrity.

The module establishes the methodological conditions required to ensure that every operational interaction with a tool, service, API, component, device, or external capability remains continuously governed rather than being assumed to be trustworthy.

A tool does not become trustworthy because it exists.

A tool becomes trustworthy only when its interactions remain governed.

TIM governs those interactions.

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## Module Operational Role

TIM defines the tool interaction governance layer of Runtime Integrity Standard™.

Its role is to ensure that operational systems maintain trustworthy, observable, and governable interactions with every tool involved in execution.

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## Module Operational Space

TIM governs:

- tool interaction integrity
- tool identification
- interaction authorization
- interaction traceability
- interaction verification
- interaction control
- external service governance
- operational interaction evidence

The module applies wherever operational systems interact with internal or external tools, services, APIs, platforms, devices, software components, or intelligent capabilities.

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## Module Function

The module applies wherever organizations must preserve:

- identifiable tool interactions
- authorized tool usage
- controlled operational exchanges
- verifiable interaction results
- reconstructable interaction history
- trustworthy governance evidence

Its function is to ensure that every operational interaction can be independently verified as governance-valid throughout its complete lifecycle.

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## Minimum Implementation Framework

### 1. Define the Tool Interaction Object

Identify which tools, services, APIs, devices, platforms, or operational components require governance oversight.

### 2. Define Tool Interaction Integrity Conditions

The organization shall define:

- approved tool identities
- authorized interaction conditions
- interaction boundaries
- verification requirements
- governance evidence
- interaction traceability

## 3. Define Interaction Degradation Detection Logic

The governance methodology shall identify:

- unknown tools
- unauthorized interactions
- unverifiable outputs
- missing interaction evidence
- uncontrolled tool behavior
- governance-invalid interactions

## 4. Define Operational Response or Governance Logic

Governance response may include:

- interaction validation
- interaction suspension
- tool isolation
- governance escalation
- operational review
- interaction recovery

## 5. Preserve Traceability & Restrict Invalid Conditions

The organization shall preserve reconstructable traceability of:

- tool identity
- interaction history
- authorization records
- operational evidence
- governance interventions
- resulting operational outcomes

Operational integrity shall not be considered valid if materially significant tool interactions cannot be reconstructed, reviewed, verified, or governed.

Use Case 1 — AI Agent Using External APIs

### Scenario

An AI agent performs operational tasks by interacting with multiple external APIs.

### Application

TIM ensures that every API interaction remains identifiable, authorized, traceable, verifiable, and continuously governed.

### Result

The organization strengthens operational transparency, governance confidence, and runtime integrity.

Use Case 2 — Industrial Automation Platform

### Scenario

A manufacturing platform coordinates robots, sensors, quality systems, and production software through multiple operational tools.

### Application

TIM provides the methodological governance framework required to preserve trustworthy interactions across the complete operational ecosystem.

### Result

The organization improves operational resilience, governance consistency, and confidence in system interactions.

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## Canonical Closing Statement

Operational trust is not created by the tools we use, but by the integrity of every interaction between them.

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# Related Documents

→ Runtime Integrity Standard™ (RIS™)

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# Canonical Source

https://originopenfoundation.org/